

Homework 3

Question 1.

- (a) Show that for any $X \in \mathbb{R}^{n \times n}$, the series

$$I + X + \frac{X^2}{2} + \cdots + \frac{X^k}{k!} + \cdots$$

is always convergent. The limit of this series is usually denoted as $\exp(X)$ and called matrix exponential of X .

- (b) Show that the set of invertible matrices

$$\text{GL}(n) := \{X \in \mathbb{R}^{n \times n} : \det(X) \neq 0\}$$

is an open set in $\mathbb{R}^{n \times n}$.

- (c) Show that the set of nilpotent matrices $\{X \in \mathbb{R}^{n \times n} : X^k = 0 \text{ for some } k \in \mathbb{N} \cup \{0\}\}$ is a closed set in $\mathbb{R}^{n \times n}$.
- (d) Show that the set of orthogonal matrices

$$\text{O}(n) := \{X \in \mathbb{R}^{n \times n} : X^\top X = I\}$$

is a compact set in $\mathbb{R}^{n \times n}$.

- (e) Show that the set of symmetric positive definite matrices

$$\mathbb{S}_{++}^n := \{X \in \mathbb{S}^n : \mathbf{y}^\top X \mathbf{y} > 0 \text{ for all } \mathbf{0} \neq \mathbf{y} \in \mathbb{R}^n\}$$

is an open set in $\mathbb{S}^n := \{X \in \mathbb{R}^{n \times n} : X^\top = X\}$ and that its closure is the set of symmetric positive semidefinite matrices

$$\mathbb{S}_+^n := \{X \in \mathbb{S}^n : \mathbf{y}^\top X \mathbf{y} \geq 0 \text{ for all } \mathbf{y} \in \mathbb{R}^n\}.$$

Question 2.

- (a) Let $f : \Omega \rightarrow \mathbb{R}$ where $\Omega \subseteq \mathbb{R}^{m \times n}$ is unbounded. Show that all sublevel sets of f are bounded if and only if

$$\lim_{k \rightarrow \infty} \|X_k\| = +\infty \implies \lim_{k \rightarrow \infty} f(X_k) = +\infty$$

assuming $X_k \in \Omega$ for all $k \in \mathbb{N}$. Deduce that if such an f is continuous and Ω is closed, then f has a global minimizer $X_* \in \Omega$.

(b) Let $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be in $C^1(\mathbb{R}^{m \times n})$. Suppose f satisfies

$$\left| \frac{\partial f}{\partial x_{ij}}(X) \right| \leq K, \quad i = 1, \dots, m, j = 1, \dots, n,$$

for all $X \in \mathbb{R}^{m \times n}$ where $K > 0$ is a constant. Show that

$$|f(X) - f(Y)| \leq \sqrt{mn}K\|X - Y\|$$

for all $X, Y \in \mathbb{R}^{m \times n}$ (Hint: Apply the univariate mean-value theorem mn times).

(c) Define $f : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}$ by

$$f \begin{pmatrix} x & y \\ z & t \end{pmatrix} = \begin{cases} \left[1 - \cos \left(\frac{x^2 + y^2 + z^2}{t} \right) \right] \sqrt{x^2 + y^2 + z^2 + t^2} & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases}$$

Show that f is continuous but not differentiable at $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. For which $V \in \mathbb{R}^{2 \times 2}$ does the directional derivative $\partial_V f \left(\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \right)$ exist?

Question 3.

Let $A : \mathbb{R} \rightarrow \mathbb{R}^{n \times n}$ be the matrix-valued function

$$A(x) = \begin{bmatrix} a_{11}(x) & \cdots & a_{1n}(x) \\ \vdots & & \vdots \\ a_{n1}(x) & \cdots & a_{nn}(x) \end{bmatrix}$$

where the functions $a_{ij} : \mathbb{R} \rightarrow \mathbb{R}$ are all in $C^1(\mathbb{R})$ for all $i, j = 1, \dots, n$.

(a) Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \text{tr}(A(x)^3)$ is differentiable and that

$$f'(x) = 3 \text{tr}(A(x)^2 A'(x))$$

where

$$A'(x) = \begin{bmatrix} a'_{11}(x) & \cdots & a'_{1n}(x) \\ \vdots & & \vdots \\ a'_{n1}(x) & \cdots & a'_{nn}(x) \end{bmatrix}$$

(b) Let $n = 2$. Suppose $A(x) \in \mathbb{S}_{++}^2$ for all $x \in \mathbb{R}$ and define $B : \mathbb{R} \rightarrow \mathbb{S}^{2 \times 2}$ by $B(x) = A(x)^{-1}$. Show that

$$\frac{d}{dx} \log \det A(x) = \sum_{i,j=1}^2 a'_{ij}(x) b_{ij}(x)$$

This is actually true for arbitrary n . For $n = 1$, it reduces to $(\log a(x))' = a'(x)/a(x)$.